

5.1 Relation



Introduction

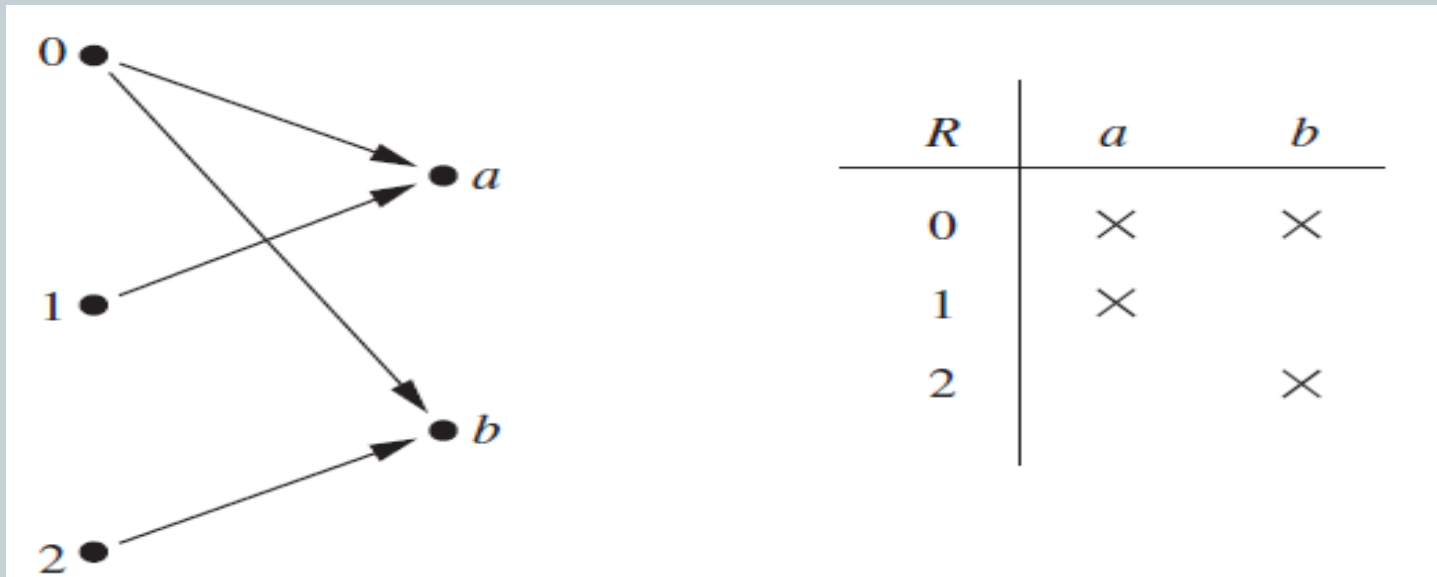


- The most direct way to express a relationship between elements of two sets is to use ordered pairs made up of two related elements. For this reason, sets of ordered pairs are called binary relations.
- Relations are subsets of two given sets.
- Let A and B be sets. A *binary relation from A to B* is a subset of $A \times B$. i.e $R \subseteq A \times B$.
- In other words, a binary relation from A to B is a set R of ordered pairs where the first element of each ordered pair comes from A and the second element comes from B . We use the notation $a R b$ to denote that $(a, b) \in R$ and $a \not R b$ to denote that $(a, b) \notin R$. Moreover, when (a, b) belongs to R , a is said to be **related to** b by R .



Example:

- Let $A = \{0, 1, 2\}$ and $B = \{a, b\}$. Then $\{(0, a), (0, b), (1, a), (2, b)\}$ is a relation from A to B . This means, for instance, that $0 R a$, but that $1 \not R b$. Relations can be represented graphically, as shown in Figure , using arrows to represent ordered pairs. Another way to represent this relation is to use a table



Domain and Range



- Suppose R is a relation from A to B then, the domain of relation R is the set of all first elements of ordered pair which belongs to R . it is denoted by $\text{Dom}(R)$.
- The range of relation R is the set of all second elements of ordered pairs which belongs to R . it is denoted by $\text{range}(R)$.
- Eg: $R = \{(1,2), (1,4), (2,4)\}$
- $\text{Dom}(R) = \{1,2\}$
- $\text{Range}(R) = \{2,4\}$

Relations on a Set



- A relation on a set A is a relation from A to A .
- In other words, a relation on a set A is a subset of $A \times A$.

Example:

Let A be the set $\{1, 2, 3, 4\}$. Which ordered pairs are in the relation $R = \{(a, b) \mid a \text{ divides } b\}$?

Solution:

Because (a, b) is in R if and only if a and b are positive integers not exceeding 4 such that a divides b , we see that

$$R = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 4), (3, 3), (4, 4)\}.$$

Properties of Relations



1. Reflexive relation: A relation R on a set A is called reflexive if $(a, a) \in R$ for every element $a \in A$. and is irreflexive if $(a, a) \notin R$

We see that a relation on A is reflexive if every element of A is related to itself.

Example:

Consider the following relations on $\{1, 2, 3, 4\}$:

$R1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\},$

$R2 = \{(1, 1), (1, 2), (2, 1)\},$

$R3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\},$

$R4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\},$

$R5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\},$

$R6 = \{(3, 4)\}.$

Which of these relations are reflexive?

Properties of Relations



Solution:

The relations R_3 and R_5 are reflexive because they both contain all pairs of the form (a, a) , namely, $(1, 1)$, $(2, 2)$, $(3, 3)$, and $(4, 4)$. The other relations are not reflexive because they do not contain all of these ordered pairs.

In particular, R_1 , R_2 , R_4 , and R_6 are not reflexive i.e. irreflexive because (a, a) is not in any of these relations.

Properties of Relations



Consider these relations on the set of integers:

$$R_1 = \{(a, b) \mid a \leq b\},$$

$$R_2 = \{(a, b) \mid a > b\},$$

$$R_3 = \{(a, b) \mid a = b \text{ or } a = -b\},$$

$$R_4 = \{(a, b) \mid a = b\},$$

$$R_5 = \{(a, b) \mid a = b + 1\},$$

$$R_6 = \{(a, b) \mid a + b \leq 3\}.$$

Which of the relations are reflexive?

Properties of Relations



The reflexive relations are R_1 (because $a \leq a$ for every integer a), R_3 , and R_4 . For each of the other relations in this example it is easy to find a pair of the form (a, a) that is not in the relation.

2.Symmetric relation: A relation R on a set A is called symmetric if $(b, a) \in R$ whenever $(a, b) \in R$, for all $a, b \in A$.

3.Antisymmetric relation: A relation R of a set A is antisymmetric if $(a,b) \in R$ and $(b,a) \in R$, then $a=b$. If $a \neq b$, then $(b, a) \notin R$ or $(a, b) \notin R$.

4.Asymmetric relation: A relation R on a set A is asymmetric if $(a,b) \in R$ but $(b,a) \notin R$.



Concept of a

Example 7 Consider the following relations on $\{1, 2, 3, 4\}$:

$$R_1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\},$$

$$R_2 = \{(1, 1), (1, 2), (2, 1)\},$$

$$R_3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\},$$

$$R_4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\},$$

$$R_5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\},$$

$$R_6 = \{(3, 4)\}.$$

Which of these relations are reflexive?

Solution The relations R_3 and R_5 are reflexive because they both contain all pairs of the form (a, a) , namely, $(1, 1)$, $(2, 2)$, $(3, 3)$, and $(4, 4)$. The other relations are not reflexive because they do not contain all of these ordered pairs. In particular, R_1 , R_2 , R_4 , and R_6 are not reflexive because $(3, 3)$ is not in any of these relations. ◀



Which of the relations are symmetric and which are antisymmetric?

Solution:

The relations R_2 and R_3 are symmetric, because in each case (b, a) belongs to the relation whenever (a, b) does. For R_2 , the only thing to check is that both $(2, 1)$ and $(1, 2)$ are in the relation. For R_3 , it is necessary to check that both $(1, 2)$ and $(2, 1)$ belong to the relation, and $(1, 4)$ and $(4, 1)$ belong to the relation. The reader should verify that none of the other relations is symmetric.

R_4 , R_5 , and R_6 are all antisymmetric. For each of these relations there is no pair of elements a and b with $a \neq b$ such that both (a, b) and (b, a) belong to the relation.



5. Transitive relation: A relation R on a set A is called transitive if whenever $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$, for all $a, b, c \in A$.

6. Equivalence relation: A relation R on a set A is called equivalence if it is reflexive, symmetric and transitive.



Example:

Let $A = \{1, 2, 3, 4, 5\}$. Show that a relation R is equivalence such that $R = \{(1,1), (1,5), (2,2), (2,4), (3,3), (4,2), (4,4), (5,1), (5,5)\}$

Soln: To prove R is equivalence we need to show that R satisfies reflexive, symmetric and transitive property.

Now, in R , $\forall 1, 2, 3, 4, 5 \in A$

There exist $(1,1) \in R$, $(2,2) \in R$, $(3,3) \in R$, $(4,4) \in R$ and $(5,5) \in R$ so this relation is reflexive.

Again, in R ,

There exist $(1,5) \in R$ then $(5,1) \in R$ and $(2,4) \in R$ then $(4,2) \in R$ which makes the relation symmetric.

Again, in R

There exist $(1,5) \in R$ and $(5,1) \in R$ then $(1,1) \in R$

$(2,4) \in R$ and $(4,2) \in R$ then $(2,2) \in R$ which makes the relation transitive.

Since R satisfies reflexive, symmetric and transitive it is equivalence.

Combining Relations



Because relations from A to B are subsets of $A \times B$, two relations from A to B can be combined in any way two sets can be combined.

Example:

Let $A = \{1, 2, 3\}$ and $B = \{1, 2, 3, 4\}$. The relations $R1 = \{(1, 1), (2, 2), (3, 3)\}$ and $R2 = \{(1, 1), (1, 2), (1, 3), (1, 4)\}$ can be combined to obtain

$$R1 \cup R2 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (3, 3)\},$$

$$R1 \cap R2 = \{(1, 1)\},$$

$$R1 - R2 = \{(2, 2), (3, 3)\},$$

$$R2 - R1 = \{(1, 2), (1, 3), (1, 4)\}.$$

Types of Relation:



Complementary Relation: let R be a relation from a set A to B . the complementary relation of R consists of those ordered pairs which are not in R . It is denoted by $\overline{R}$.

- Let R be a relation on set $A=\{1,2,3\}$ defined as $R=\{(x, y): x<y\}$. Find the complement of R .

Soln: $A*A=\{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3)\}$

$R=\{(1,2), (1,3), (2,3)\}$

Now, $\overline{R}=\{(1,1), (2,1), (2,2), (3,1), (3,2), (3,3)\}$



Inverse Relation: let R be a relation from A to B then inverse of R is the relation from B to A .i.e. it consists of those ordered pairs which when reversed belongs to R . it is denoted by R^{-1} .

Example: let $A=\{1,2,3,4\}$ and $B=\{a, b\}$ and $R=\{(1,a), (1,b), (2,a), (3,b), (4,b)\}$

Then $R^{-1} = \{(a,1), (b,1), (a,2), (b,3), (b,4)\}$

Identity Relation: A relation R from A to A is said to be identity relation. It is denoted by I_A

$I_A = \{(x, x): x \in A\}$

n-ary Relations



- The relationship among more than two sets are called *n*-ary relation.
- Let $A_1, A_2, \dots, A_n$ be sets. An *n*-ary relation on these sets is a subset of $A_1 \times A_2 \times \dots \times A_n$.

Example:

Let R be the relation on $N \times N \times N$ consisting of triples (a, b, c) , where a, b , and c are integers with $a < b < c$. Then $(1, 2, 3) \in R$, but $(2, 4, 3) \notin R$.

Representation of Relations:



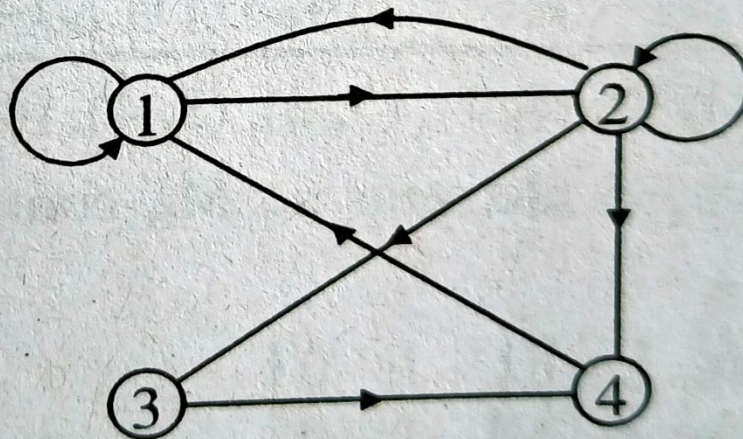
Directed graph of relations: for graphical representation of relation on a set, each element of the set is represented by a node.

An arc is drawn from each node to its related node.

example: let $A = \{1, 2, 3, 4\}$ and $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (2, 3), (2, 4), (3, 4), (4, 1)\}$

Then the diagraph of relation R is,

Then the diagraph of R is as shown in figure.





Matrix of Relation: let $A = \{a_1, a_2, \dots, a_m\}$ and $B = \{b_1, b_2, \dots, b_n\}$ be finite sets containing m and n elements respectively and let R be relation from A to B then R can be represented by mn matrix $M_R = [M_{ij}]_{mn}$

- $M_{ij} = \begin{cases} 1 & \text{if } (a_i, b_j) \in R \\ 0 & \text{if } (a_i, b_j) \notin R \end{cases}$

- Then the matrix M_R is called the matrix of relation R .

Complement of matrix: The complement of matrix M_R is obtained by changing 0's to 1's and 1's to 0's.

Transpose of matrix: The transpose of M_R is obtained by interchanging its rows and columns.



Let R be a relation from the set $A = \{1,2,3\}$ on $R = \{(1,1), (1,3), (2,3), (3,3)\}$ then find M_R .

Soln:

$A * A = \{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3)\}$

Since set A contain 3 elements M_R contain 3 rows and 3 columns.

•
$$M_R = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} \end{matrix}$$

Boolean matrix



- The matrix with 0's or 1's entries is called boolean matrix

- $a_1 \wedge a_2 = \begin{cases} 1 & \text{if } a_1 = a_2 = 1 \\ 0 & \text{otherwise} \end{cases}$

- $a_1 \vee a_2 = \begin{cases} 1 & \text{if } a_1 = 1 \text{ or } a_2 = 1 \\ 0 & \text{otherwise} \end{cases}$

Example: find $A \wedge B$ and $A \vee B$ of the Boolean Matrix.

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} \quad A \vee B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad A \wedge B = \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}$$

Boolean Product



Find the Boolean Product of given matrix:

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$A.B = \begin{pmatrix} (1 \wedge 0) \vee (0 \wedge 1) \vee (1 \wedge 0) & (1 \wedge 1) \vee (0 \wedge 1) \vee (1 \wedge 0) & (1 \wedge 0) \vee (0 \wedge 1) \vee (1 \wedge 1) \\ (1 \wedge 0) \vee (1 \wedge 1) \vee (0 \wedge 0) & (1 \wedge 1) \vee (1 \wedge 1) \vee (0 \wedge 0) & (1 \wedge 0) \vee (1 \wedge 1) \vee (0 \wedge 1) \\ (0 \wedge 0) \vee (0 \wedge 1) \vee (1 \wedge 0) & (0 \wedge 1) \vee (0 \wedge 1) \vee (1 \wedge 0) & (0 \wedge 0) \vee (0 \wedge 1) \vee (1 \wedge 1) \end{pmatrix}$$

$$A.B = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

Partition of set



Let A be a set and $A_1, A_2, \dots, A_n$ be its non-empty subsets. The subsets form a partition P of A if

- The union of subsets in P is equal to A .
- The partition P does not contain the empty set.
- The intersection of any distinct subsets in P is empty.

Example: which of the following collection of subsets are partitions of $A = \{0, 1, 2, 3, 4, 5\}$

1. $\{0, 1, 2\}, \{4, 3\}, \{4, 5\}$
2. $\{0, 1, 2\}, \{4, 3, 5\}, \{\}$
3. $\{5, 4, 0, 3\}, \{2\}, \{1\}$
4. $\{5\}, \{4, 3\}, \{0, 2\}$
5. $\{1\}, \{2\}, \{5\}, \{0\}, \{4\}, \{3\}$

Partition of set



1. is not a partition of set A since the element 4 is contained in two subsets.
2. is not partition because they have empty set.
3. is a partition of A.
4. is not a partition of A because 1 is missing.
5. is a partition of set A

Equivalence Class:

An equivalence class of a relation R on a set A for each element X of A, is denoted by $[X]_R$ and is defined as

$$[X]_R = \{ y \mid y \in A \text{ \& } (X, Y) \in R \}$$

Note: for finding equivalence class the relation must be equivalent.

Equivalence Class



Example:

$$A = \{1, 2, 3, 4, 5\}$$

$$R = \{(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (1, 2), (2, 1), (4, 5), (5, 4)\}$$

Sol:

$$\text{equivalence class of 1 : } [1]_R = \{1, 2\}$$

$$\text{equivalence class of 2 : } [2]_R = \{1, 2\}$$

$$\text{equivalence class of 3 : } [3]_R = \{3\}$$

$$\text{equivalence class of 4 : } [4]_R = \{4, 5\}$$

$$\text{equivalence class of 5 : } [5]_R = \{4, 5\}$$

Composition of Relation



- Let A , B & C be sets and R be a relation from A to B and S be a relation from B to C i.e. R is a subset of $A \times B$ and S is a subset of $B \times C$.
- Then R and S give rise to a relation from A to C indicated by SoR and defined as:

$$SoR = \{ (a,c) \mid b \in B \text{ for which } (a,b) \in R \text{ and } (b, c) \in S \}$$

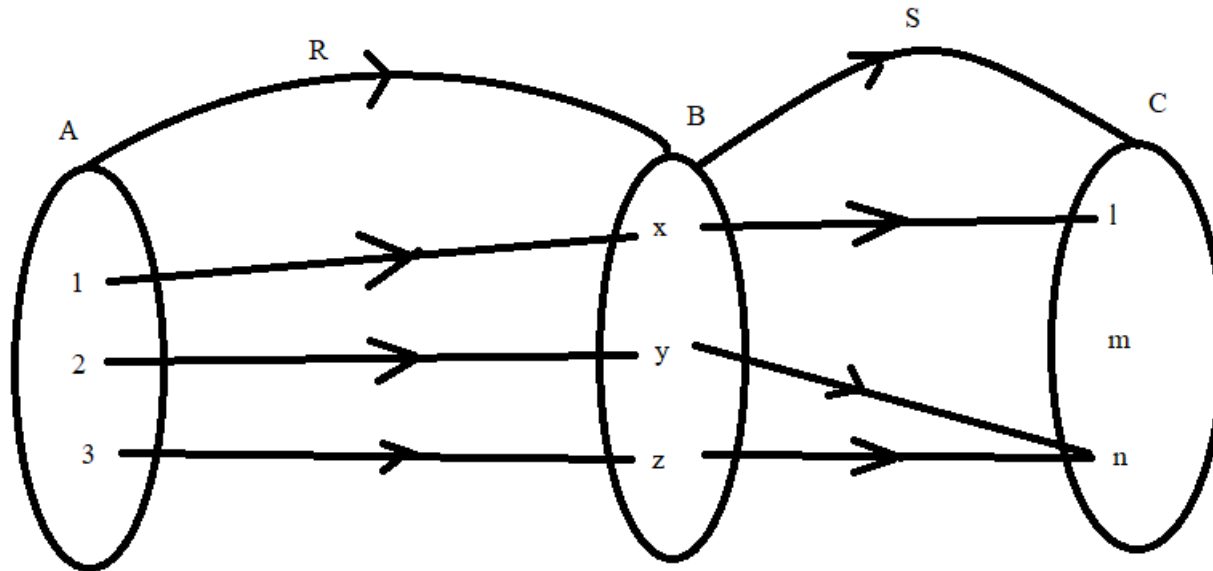
Example: $A = \{1, 2, 3\}$, $B = \{x, y, z\}$, $C = \{l, m, n\}$

$R: A \rightarrow B \{(1,x), (2,y), (3,z)\}$

$S: B \rightarrow C \{(x, l), (y, n), (z, n)\}$

The composition of R and S is SoR

Composition of Relation



From the above diagram,

$1 \rightarrow x \rightarrow l \quad :: (1, l) \in \text{SoR}$

$2 \rightarrow y \rightarrow n \quad :: (2, n) \in \text{SoR}$

$3 \rightarrow z \rightarrow n \quad :: (3, n) \in \text{SoR}$

$\text{SoR} = \{ (1, l), (2, n), (3, n) \}$

Partial order relation



A relation R on a set A is said to be partial order relation if it is reflexive, antisymmetric and transitive.

Example: let $A = \{1, 2, 3, 4\}$

$R = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,3), (2,4), (3,3), (3,4), (4,4)\}$

Determine (A, R) is a poset?

Soln: To determine (A, R) a poset we have to show the R is reflexive, antisymmetric and transitive.

For reflexive: $(a, a) \in R \forall a \in A$

$(1,1) \in R$

$(2,2) \in R$

$(3,3) \in R$

$(4,4) \in R$

So R is reflexive.

Partial order relation



For antisymmetric: if $(a, b) \in R$ and $(b, a) \in R$, then $a=b$. If $a \neq b$, then $(b, a) \notin R$ or $(a, b) \notin R$.

$(1,2) \in R$ but $(2,1) \notin R$

$(1,3) \in R$ but $(2,1) \notin R$

$(1,4) \in R$ but $(2,1) \notin R$

$(2,3) \in R$ but $(2,1) \notin R$

$(2,4) \in R$ but $(2,1) \notin R$

$(3,4) \in R$ but $(2,1) \notin R$

So R is antisymmetric.

For transitive: $(a, b) \in R$ and $(b, c) \in R$ then $(a, c) \in R$

$(2,3) \in R$ and $(3,4) \in R \rightarrow (2,4) \in R$

$(1,3) \in R$ and $(3,4) \in R \rightarrow (1,4) \in R$

$(1,3) \in R$ and $(3,3) \in R \rightarrow (1,3) \in R$

So R is transitive

$\therefore$ Hence (A, R) is poset.